sanas South African National Accreditation System	**TECHNICAL AND PRODUCTION REQUIREMENTS of ISO 17034:2016**	Organisation's SANAS No/s. **Document Review**

Date/s of evaluation	**March 2026**
Assessor/s & Observers	**Phuti Mathopa**
Laboratory	**Azuresky Pty Ltd**
Area / field of operation	**RMP**
Laboratory Representative	

This report covers the following:

Document Review only	X	Implementation on Site Visit only		Document Review and Site Visit		Other	

REQUIREMENTS & COMMENTS. ***Compliance = C, Non-compliance = NC***

NB1: *References to ISO/IEC 17034:2016 are in italics. The order of assessment need not follow the order of the checklist. Assessors are expected to know & have the standard, this worksheet is designed as guidance to prompt detailed recording of the process.*

REFER TO ISO 17034:2016 FOR DETAIL AND FOR CLARIFICATION NOTES.

NOTE 1: *For **CAB's comments**: The CAB must provide information on how requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*

NOTE 2: *For **Assessor's Comments**: The Assessor must provide information on the CAB's conformity with the requirements*

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<table>
<tr><th>CLAUSE</th><th>ISO/IEC 17034:2016 REQUIREMENTS</th><th>CAB's COMMENTS</th><th>C/ NC</th><th>ASSESSOR's COMMENTS</th></tr>
<tr><td colspan="5">7. TECHNICAL AND PRODUCTION REQUIREMENTS</td></tr>
<tr><td>7.1</td><td colspan="4">General requirements</td></tr>
<tr><td></td><td colspan="4">A CRM has at least one certified value
Clause 7.9 applies only to certified values</td></tr>
<tr><td>7.2</td><td colspan="4">Production planning:</td></tr>
<tr><td>7.2.1</td><td>• Has the RMP identified those process that directly affect the quality of the RM?</td><td></td><td>C</td><td rowspan="3">Document, Production Plan, Ref: ASMF7.2.1, details that homogeneity testing is conducted on sample/material product for Quick Ash%.
The samples are retained for determination of stability test for a period of 4 (four) weeks.
The material retained after the PT production is packaged for Certified Reference Material production.
PT and RM samples that have achieved sufficient homogeneity and stability requirements are used for RMP
Table below indicates the process/plan for RMP
<table><tr><td>Field</td><td>Field</td></tr><tr><td>Entry</td><td>Entry</td></tr><tr><td>CRM Name</td><td>CRM Name</td></tr><tr><td>GA CRM AS002</td><td>GA CRM AS002</td></tr><tr><td>Unique batch Number</td><td>Unique batch Number</td></tr><tr><td>CRM AS002 2025-05</td><td>CRM AS002 2025-05</td></tr><tr><td>Intended Use</td><td>Intended Use</td></tr><tr><td>Calibration, Control Sample, and Training of staff</td><td>Calibration, Control Sample, and Training of staff</td></tr></table></td></tr>
<tr><td></td><td>• Has the RMP planned those process that directly affect the quality of the RM?</td><td></td><td>C</td></tr>
<tr><td></td><td>• Has the production plan been documented?</td><td></td><td>C</td></tr>
<tr><td>7.2.2</td><td>• Has any technical input from sub-contractors been specified?</td><td></td><td>NA</td><td>No subcontracting has been done</td></tr>
</table>

<table>
<tr><th>CLAUSE</th><th>ISO/IEC 17034:2016 REQUIREMENTS</th><th>CAB's COMMENTS</th><th>C/ NC</th><th>ASSESSOR's COMMENTS</th></tr>
<tr><td></td><td>• Is this technical input documented, and regularly reviewed?</td><td></td><td>C</td><td>Contractual Matters - Review of Contracts, Tenders, and Customer Requests Checklist details how sub - contractors are handled, Ref: ASMF4.2.1</td></tr>
<tr><td>7.2.3</td><td>Has the RMP addressed, during the planning sage:-</td><td></td><td>NA</td><td></td></tr>
<tr><td>a)</td><td>Material selection? (including sampling, where applicable)</td><td></td><td>C</td><td>Addressed in section 1 of document ASMF7.2.1 - A list of materials and equipment to be used in the production of the GA CRM AS002 are tabled and listed</td></tr>
<tr><td>b)</td><td>Verification of the identity of the material?</td><td></td><td></td><td></td></tr>
<tr><td>c)</td><td>Maintaining a suitable environment for production?</td><td></td><td>C</td><td>Temperature and humidity are checked before starting with work and closely monitor these throughout the shift.
<table>
<tr><th>Description</th><th>Equipment used</th><th>Environmental Conditions</th></tr>
<tr><td>Air drying</td><td>-</td><td>Atmospheric temperature</td></tr>
<tr><td>Stage Crushing</td><td>25,10,3 mm Crushers</td><td>Room temperature</td></tr>
<tr><td>Pulverise</td><td>Serial no: 22L90713190645</td><td>Room temperature</td></tr>
<tr><td>Homogeneity Drum</td><td>-</td><td>-</td></tr>
<tr><td>Homogeneity Validation (analysis)</td><td>ISO Ash Furnace</td><td>19-26 °C</td></tr>
<tr><td>Packaging</td><td>±150g</td><td>Room temperature</td></tr>
<tr><td>Storage</td><td>Storage room</td><td>≤ 20°C</td></tr>
</table></td></tr>
<tr><td>d)</td><td>Material processing?</td><td></td><td>C</td><td>Defined in section 3 of Production plan ASMF7.2.1</td></tr>
</table>

<table>
<tr><th>CLAUSE</th><th>ISO/IEC 17034:2016 REQUIREMENTS</th><th>CAB's COMMENTS</th><th>C/ NC</th><th>ASSESSOR's COMMENTS</th></tr>
<tr><td>e)</td><td>Choice of measurement procedures?</td><td></td><td>C</td><td rowspan="2">The following authorised procedures are to be used:
- AS002IW: Conditioning of samples
- AS003IW: Determination of Ash content
- AS006IW: Determination of Volatile Matter
- AS004IW: Determination of Inherent Moisture</td></tr>
<tr><td>f)</td><td>Validation of the measurement procedures?</td><td></td><td>C</td></tr>
<tr><td>g)</td><td>Calibration/verification of measuring equipment?</td><td>.</td><td>C</td><td>Method Validation reports for the Determination Ash, Ref: AS003MV, Calorific Value, Ref: AS002MV; Inherent Moisture, Ref: AS004MV; ISO Volatile Matter: Ref: ASMV 001

Defined in respective testing method for coal i.e. CV test method, Ash test method, VM test method, IM test method and sample preparation method.
Implementation to be verified during Initial Assessment</td></tr>
<tr><td>h)</td><td>Specification of acceptance criteria? (homogeneity including sampling)</td><td></td><td>C</td><td>Defined in section 3 and 4 of Production plan ASMF7.2.1
<table>
<tr><th>Description</th><th>Equipment used</th><th>Environmental Conditions</th><th>Acceptance criteria</th></tr>
<tr><td>Air drying</td><td>-</td><td>Atmospheric temperature</td><td>Sample should be visibly dry</td></tr>
<tr><td>Stage Crushing</td><td>25,10,3 mm Crushers</td><td>Room temperature</td><td>Crusher Efficiency ≥90%</td></tr>
<tr><td>Pulverise</td><td>Serial no: 22L90713190645</td><td>Room temperature</td><td>Pulveriser efficiency ≥95%</td></tr>
<tr><td>Homogeneity Drum</td><td>-</td><td>-</td><td>~ 4 hours</td></tr>
<tr><td>Homogeneity Validation (analysis)</td><td>ISO Ash Furnace</td><td>19-26 °C</td><td>ANOVA, SSbetween<Check Value</td></tr>
<tr><td>Packaging</td><td>±150g</td><td>Room temperature</td><td>-</td></tr>
<tr><td>Storage</td><td>Storage room</td><td>≤ 20°C</td><td>Temperature <23°C</td></tr>
</table>
Implementation to be verified during Initial Assessment</td></tr>
</table>

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i)	Specification of acceptance criteria? (stability including sampling)		C	Defined in section 3 and 4 of Production plan ASMF7.2.1
j)	Design and organization of characterization? (sampling)		C	Defined in section 4 of ASMF 7.12, Characterisation .The lab uses the following approaches: • One laboratory using validated methods • Multiple laboratories using the same or different validated methods • Combination of methods or historical data, where justified.
k)	Commutability? (where appropriate)		NC	It does not seem like commutability is understood and thus not well addressed Commutability is not about distribution and transportation The lab to review this clause and address accordingly
l)	Assigning property values?		C	Detailed in procedure ASMF 7.13, section 5 Property values are assigned e.g., Ash content in % on dry basis and units.
m)	Establishing uncertainty budgets incl. uncertainty of certified values?		C	Defined in document ASMF UB001 Implementation to be verified during Initial Assessment
n)	Defined acceptance criteria for measurement levels and their uncertainties?		C	Defined in document ASMF UB001 Implementation to be verified during Initial Assessment

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<tr><td>o)</td><td>Metrological traceability of measurement results and certified values?</td><td></td><td>C</td><td>Defined in ASMF 7.9
Implementation to be verified during Initial Assessment</td></tr>
<tr><td>p)</td><td>Issuing of RM documents?</td><td></td><td>C</td><td>A soft copy of the reference material certificate is emailed to the client, and a hard copy accompanies the purchased CRM</td></tr>
<tr><td>q)</td><td>Ensuring storage facilities and conditions?</td><td></td><td>C</td><td>Defined in procedure ASMF7.2.3, section 5 which states the parameter of control, the range to apply the control measure and the respective areas of the laboratory.
<table><tr><td>Parameter</td><td>Acceptable Range</td><td>Controlled Areas</td></tr><tr><td>Temperature</td><td>19-26 °C</td><td>All preparation, weighing, and packaging</td></tr><tr><td>Humidity</td><td>25-45 % RH (unless specified)</td><td>Drying, crushing, packaging, analysis.</td></tr><tr><td>Particulate</td><td>Minimal visible particles</td><td>Cleanroom or clean bench</td></tr><tr><td>Airflow</td><td>Positive pressure if required</td><td>Clean areas</td></tr><tr><td>Light</td><td>Avoid direct sunlight</td><td>Storage and packaging</td></tr></table>Implementation to be verified during Initial Assessment</td></tr>
<tr><td>r)</td><td>Appropriate labelling and packaging?</td><td></td><td>C</td><td>Defined in procedure ASMF7.15, Distribution And Customer Support Procedure</td></tr>
<tr><td>s)</td><td>Transport arrangements?</td><td></td><td>C</td><td>Defined in procedure ASMF7.15, Distribution And Customer Support Procedure</td></tr>
</table>

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t)	Post production stability monitoring?		C	The CAB addresses stability monitoring of certified reference materials post production in 7.4 Material Handling and Storage under section 6 post production stability monitoring. This section outlines the study, how it is conducted and future amendments (where applicable) as a reaction to the outcome of the study. See attached Stability Monitoring document as evidence.
u)	Adequacy of post distribution service for RM users?		C	Where issues arise from the distributions of such materials produced in the lab, clients log a complaint and the lab has 7.18 Complaints and handling procedure to combat any issues that may arise from any of the services the CAB offers. The complaints and appeals procedure is displayed publicly in the CABs website for clients to access when issues arise.
7.2.4	Where multiple batches of RM with equivalent properties are produced, has the RMP verified that information from previous batches is still applicable to the new batch?		C	The lab outlined in AS001IW Procedure On Sample Preparation Of Hard Coal Rev 02 under 4.1 Rationale , how the assignment of batches to originally produced certified reference materials will be conducted when the need arises.
7.3	**Production Control**			
	• Has the RMP verified that the production plan has been implemented as specified?		C	The checklist used during sample preparation, testing and LIMS data is able to provide auditable data for assessment per CRM produced
	• Have deviations from the plan been documented and approved?		C	Procedure ASMF 8.7.3 addresses deviations and improvements
7.4	**Material handling and storage**			

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7.4.1	• Has the RMP made arrangements to ensure the integrity of candidate RM and RM's throughout the process?		C	Material handling and storage ASMF 7.4.1 designed and implemented to maintain stability, quality, identity and traceability throughout the handling processes According to Procedure ASMF 7.2.3 is implemented and the recording of condition on AS018TF checklist and AS023TF checklist, there's condition monitoring measures applied during testing and storage The packaging of CRM's is done with plastic pulped containers that have a seal proof lid. continually applying stability checks to measure deterioration of the quality of CRM ASMF 7.15 Distribution and customer support procedure section packaging and distribution. Tamper proof packaging is a requirement under this section. Implementation to be verified during Initial Assessment After production of a Reference material, the material is assigned a shelf life of 1 year. During the course of the year, on the 25th of each month the CAB conducts a stability monitoring of certified material. The acceptability criteria is set as follows: The average of each month's monitoring must be reproducible to the original assigned value. Implementation to be verified during Initial Assessment
	• Have precautions been taken against adverse environmental conditions during processing?		C	
7.4.2	Has the RMP identified, preserved and separated candidate RM and RM's from chemicals and other samples, throughout the process?		C	
7.4.3	• Has the RMP used adequate packaging?		C	
	• Have following been considered for the packaging of the RM's: light-shielding, air-free, moisture-free or inert gas packaging?		C	
	• Has the RMP secure areas or stock rooms which will prevent damage or deterioration?		C	
7.4.4	Is the condition of RM's assessed at appropriate intervals to detect possible deterioration?		C	

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7.4.5	• Are the packaging and labelling processes controlled, and do they conform to safety and transport requirements?		C	The lab has a temperature-controlled and humidity-monitored storage facility with a clear storage procedure outlined to control the temperature within the facility at 16 to 20 degrees to prolong and prevent accelerated shelf life (deterioration). RM documents and labels, also addressed in Material handling and storage procedure Implementation to be verified during Initial Assessment
	• Has the RMP defined procedures for transport to the customer?		C	Procedure 7.15 Distribution procedure defined a procedure for distribution and customer support, which outlines packaging specifications and certificates that accompany the materials distributed
7.4.6	Has the RMP taken measures to ensure the integrity of the RM, until a seal is broken, or on initial usage?		C	CRM Name: Certified Reference Material AS007, document verified and complies with how the lab ensures the integrity of RM
7.5	**Material Processing**			
7.5.1	• Has the RMP established procedures to ensure adequate processing of RM's? • Do these procedures address at least:-		C	Document, 7.2. PRODUCTION PLANNING, Ref: ASMF7.2.1 outlines the processing steps for RM production. Each commodity has a sample preparation procedure detailing the process of preparation. Hard coal sample preparation Procedure, Ref: AS001IW
a)	Qualitative analysis to ensure verification of material type or identity?			
b)	Synthesis, purification, incubation and or transformation?			

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c)	Homogenization?			
d)	Proper handling?			
e)	Measurements for control of the material processing?			
f)	Pre-treatment, cleaning or sterilization of equipment and sample containers?		C	
g)	Stabilization of the material?		C	Covered in document, AS002IW, Conditioning of test samples
h)	Packaging of the material?		C	Air-dry the sample in an oven ≤ 40 °C for ≤4 hours. Packaging is done once validation of all analytical parameters has been successfully completed and approved.
i)	Safety precautions?		C	Reference materials are dispensed from the homogenized bulk (homogeneity drum) into sealed, tamper-evident containers under controlled conditions to prevent contamination or material alteration. Each unit is then labelled to ensure unique identification, traceability to the batch, and clear reference to property values and storage conditions. Implementation to be verified during Initial Assessment
7.5.2	Is the equipment used in the processing operated in accordance with documented procedures? NOTE: Manufacturer's instructions are considered as documented procedures		C	

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7.6	• Are all the requirements of ISO/IEC 17025 met with respect to calibration & testing?		C	The lab says that equipment with potential to affect the quality and validity of results are calibrated. The lab has developed an equipment register for calibration intervals, calibration certificates and equipment status. Testing of coal material is done following procedures that are derived from internationally recognized standards (ISO 1928, SANS 5925, ISO 589. The Lab uses certified reference materials of known value at outlined intervals to monitor any equipment drifts and if the equipment still produces acceptable, repeatable and reproducible results Implementation to be verified during initial assessment
	• Are the measurements and calibrations consistent with the accuracy of the RM and any relevant standard specifications for the measurements?		C	
7.7	**Measuring equipment:**			
	Is all the measuring equipment used in the RM production in compliance with the relevant clauses of ISO/IEC 17025?		C	The lab performs equipment verification on balance verifications prior to use to ensure that equipment is still fit for its intended purpose. Certificates of calibration are attached and verified, however, implementation to be verified during initial assessment
7.8	**Data integrity and evaluation:**			
7.8.1	Does the RMP ensure that all calculations and data transfers are subject to appropriate checks?		C	According to procedure, Data Integrity, Ref: ASMF 7.11, the lab validates all information systems (LIMS, excel spreadsheets, calculations, data transfers) to safeguard data integrity prior to use and at outlined intervals. Implementation to be verified during initial assessment The lab protects its operating systems and documents by backing up the entire system quarterly in the case of
7.8.2	Has the RMP ensured that:			
a)	• Software developed in-house is validated and found adequate for use?			
	• Off-the-self software that is adapted is validated a found adequate for use?			

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b)	• Are procedures established and implemented for protecting the integrity of data?			malware/virus attacks To protect unauthorized changes to documents, the CAB has unchangeable PDF documents at points of use and personnel have the parts of the system need to carry out their activities and only authorized personnel have access to the full system
	• Do these procedures address data entry, capture, storage, transmission and processing?			
c)	• Is the data processing equipment maintained to ensure proper functioning?			Implementation to be verified during initial assessment
	• Are the environmental and operating conditions satisfactory for the maintenance of data integrity?			Implementation to be verified during initial assessment
d)	• Procedures established and implemented for the maintenance of data security?			Implementation to be verified during initial assessment
	• Do these procedures address unauthorized access, including amendments to records?			Implementation to be verified during initial assessment
	Note: Validation could include evidence of a sound theoretical basis – by reference to appropriate literature, and certain assumptions which can be shown to apply to the data.			
7.8.3	Are all statistical procedures used appropriate for their application?		C	The assignment of certified reference values is derived from statistically evaluated consensus means obtained through PT data analysis and validated measurement results. statistical processes and procedures are derived from internationally recognized standards (ISO 13528, ISO 17034 & ISO 17043). Implementation to be verified during initial assessment
7.9	**Metrological traceability of certified values**			

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7.9.1	Has the metrological traceability for all CRM's been established in accordance with ISO/IEC 17025?		C	The lab establishes metrological traceability through the use of ISO 1928 as the reference method for Gross Calorific Value determination. The calibration of the bomb calorimeter is performed using certified benzoic acid reference material, the assigned value of which is traceable to the SI unit of energy (joule, J) through the international thermochemical scale. Measuring instruments, i.e. analytical balance, temperature sensors are calibrated by a SANAS accredited laboratories suppliers. The lab demonstrates measurement comparability through participation in coal proficiency testing schemes and the routine use of certified reference materials for calibration verification. Verified through attached equipment register and document, Ref: 6.5 Metrological traceability memorandum and method validation for one of the proximate as evidence of validation. This is evident in all of its CRM certificates, with one of it attached. The statement of conformity also has this covered, Ref: CRM Name: **Certified Reference Material AS007** CRM Code: CRM AS 007 Batch Number: 1 Date of Certification: 21-10-2025 Expiry Date: One year after opening Produced and certified by: Azuresky Pty Ltd, South Africa Covered through documents, Ref: the procedures (CV), Statement of Conformity and CRM 002 The lab uses PT participation for maintaining the method; justification for non-SI traceability is recorded.
	Is there evidence of metrological traceability of the certified value to a stated reference?		C	
7.9.2	Is the stated reference, by definition to a measurement unit (by practical realization), or a measurement procedure or a measurement standard?		C	
7.9.3	Where traceability to the SI is possible, is the RMP able to demonstrate this?		C	
7.9.4	Where traceability to the SI is not possible, is the RMP able to demonstrate traceability to an appropriate reference? These references could be i) CRM's from a competent producer ii) reference measurement procedures iii) specified methods or consensus standards (ISO/IEC 17025:2017)		C	

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7.9.5	Where traceability to a higher order reference system is required, is the RMP able to demonstrate that the measurements are calibrated with standards with metrological traceable values?			Validations, ASH, IM, VOL, available and verified.
7.9.6	Is the RMP able to provide evidence that secondary parameters having a significant influence on the certified value or its uncertainty are metrologically traceable?			Secondary parameters (e.g., oven temperature, balance mass, furnace thermocouple) are listed in the equipment register and are covered by calibration certificates; their contributions are included in the uncertainty budget.
7.10	**Assessment of homogeneity:**			
7.10.1	Has the RMP conducted the homogeneity testing of the candidate RM in its final packaging? Note: Assessment of homogeneity could include prior evidence, but experimental guide study would then be necessary. The homogeneity tests require a representative number of samples chosen at random, or systematic starting from a random start point.		C	RM's are derived from a PT sample, homogeneity assessments are conducted prior to packaging of reference materials to ensure that each reference material is packaged with the confidence that the reference material is homogenized across all the packaged pulps**.**
7.10.2	Is the RMP able to provide evidence that when multiple batches are produced there is equivalence between the batches, or has the homogeneity of each batch been determined independently?		C	
7.10.3	• Have validated measurement procedures used to the homogeneity testing?		C	Multiple batches arel only be made available after a sample goes for PT and after statistical studies conclude that the assigned values are reproducible to an already established CRM, then it can form a batch of that CRM which would mean that its homogeneity is determined independently.
	• Is the selectivity and precision of the procedures fit for purpose?		C	
7.10.4	• Has the homogeneity for each property of interest been determined independently, or		C	Lab demonstrate homogeneity and ongoing stability studies that measurement of ash content, being

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	• Can it be shown, scientifically or otherwise that measurement of one property is closely associated with other properties in the group?		C	the most stable and representative parameter of the coal matrix, adequately reflects the variability of other associated properties such as volatile matter and calorific value. This scientific association is based on established correlations observed in the material and literature evidence showing ash content as a key indicator of coal composition. The CAB maintains a stability monitoring spreadsheet, and data evaluated in nine months to confirm that no significant changes occur in the ash or related parameters. This will confirm that the selected property (ash) is suitable as a representative measurand for homogeneity assessment
7.10.5	• Has the RMP for all certified values, quantified the homogeneity as an uncertainty contribution, or shown as a negligible contribution to the uncertainty?		C	
7.11	**Assessment and monitoring of stability:**			
7.11.1	Does the RMP have:			Initial Stability Test Store the same 10 bottles used for homogeneity testing in the CRM storage room (16–20 °C; humidity monitored). 2. After 30 days, re-analyze the same 10 bottles for Ash content. 3. Calculate the mean difference (Δ) between homogeneity mean and stability mean Ongoing Stability Monitoring Conduct monthly stability monitoring on the 25th of each month for a 12-month period. 2. Use archived bottles stored under reference conditions. 3. Analyze for ash content and compare to the assigned value of the CRM. If stability issues ensue, the CAB
a)	Access to the stability of all relevant properties of a RM under proposed storage conditions?			
b)	Access to the stability of all relevant properties of a RM under proposed transport conditions?			
c)	Advice on storage and use necessary to maintain stability by the user?			
d)	• Has a scheme been selected that allows for the monitoring the long-term stability?			
	• Does this scheme allow for prompt detection of changes to the stability?			

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	• Does the scheme take into account the possible rate of change?			documents that stability is to be re-done for the month and if it still does not meet the criteria set.
e)	• Where the stability cannot be ensured has the RMP made allowance for changes in the uncertainty?			Implementation to be verified during initial assessment
	• Or, where changes to the certified value can be predicted, a means of correcting the certified value and its uncertainty?			When need arises for the CAB to re-evaluate its assigned values for an already produced CRM, the lab may choose to re-evaluate the affected CRM by redistributing the said lot to its participants as a PT, conduct a new statistical analysis to assign new values and follow established prompts to communicate the amendments to its customers.
f)	• Where repeated usage of a RM is permitted, has the RMP assessed the possible effects on the stability, and taken appropriate action?			
7.11.2	• Has the RMP conducted experimental assessment of the stability?			
	• Or does the RMP have evidence of stability from prior evaluation of similar batches?			Implementation to be verified during initial assessment
7.11.3	Where multiple batches of RM are produced, does the RMP evidence that sufficient stability testing between batches has been done, to provide confidence in the stability of all batches?			
7.12	**Characterization:**			
7.12.1	Has the RMP assigned property values only after characterization has been conducted?			Assignment of property values is only concluded after statistical analysis of data has been completed. How the CAB does this is by taking consensus values from a concluded PT round and conducts validation and uncertainties of parameters and
7.12.2	• Has the RMP defined whether the characterization is quantitative or qualitative property?			

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	• And, if quantitative whether the measurand is defined operationally, or defined independently?		C	producing a certificate Each property is defined operationally in the characterization and assignment of property values and their uncertainties (e.g., "volatile matter — ISO 562, % dry basis") and the nature (quantitative) is recorded It is defined operationally by international coal ISO standards as stated in section 7.13 Implementation to be verified during initial assessment
7.12.3	Has the RMP chosen a characterization strategy appropriate to its intended use? Note: characterization could follow one of these approaches a) single reference measurement procedure in a single lab, b) using two or more methods, in two or more laboratories, c) characterized in a network of competent laboratories, d) transfer from a closely matched RM using a single measurement procedure, or e) characterization based on mass or volume of ingredients used in the preparation.		C	
7.12.4	• Has the RMP specified the characterization study, such that the properties of interest have appropriate traceability and sufficient reliability?		C	
a)	• Has the RMP documented a measurement plan that clearly describes the tasks to be performed, and is this plan communicated to responsible personnel?			
b)	• Where CRM's are certified, has the RMP demonstrated the competence of each laboratory by using data not obtained through the material to be characterized?			
7.12.5	• When evaluating the characterization data has the RPM performed a technical evaluation of the data and documents involved?			
	• Has adherence to the measurement plan been confirmed?			
	• If there have been deviations from the measurement plan, has the RMP decided if the deviation necessitates exclusion of the data from the characterization?			

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7.13	**Assignment of property values**			
7.13.1	Does the RMP have documented procedures for the assignment of property values?			The assignment of property values follows the documented SOP 7.13 which outlines experimental design, statistics and uncertainty approach. The procedure attached addresses clause 7.13.2 a-e in its entirety Implementation to be verified during initial assessment
7.13.2	Do these procedures address:		C	
a)	• Details of the experimental designs?		C	
	• Details of the statistical techniques used?			
b)	• Policies for the treatment of outliers?			
	• Policies for the treatment and investigation of anomalous results?			
c)	• Weighing techniques for results from different laboratories, or procedures?			
d)	• The approach used to assign uncertainties to property values?			
e)	• Other factors that could influence the assignment of the property values?			

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
7.13.3	Has the RMP taken in account information on test methods, equipment, uncertainties reported, and evidence of past performance?			
7.13.4	• Have outliers been excluded without investigation of the possible reasons?			
	• Have robust techniques been used where appropriate?		C	Method validation was initially conducted on the methods selected and compatibility of instruments and their uncertainties and historic PT performance were considered during characterization and value assignment.
7.13.5	Has the RMP for all 'certified' values considered all uncertainty contributors to be included in the assigned uncertainty?		C	
7.13.6	For all 'certified' values has the RMP considered the following minimum contributors:-		C	
a)	• Characterization, including differences between multiple procedures?		C	Outliers are removed during the statistical evaluation under PT statistical analysis. To assign robust averages on CRM values and their standard deviations, the lab employs Algorithm A of ISO 13528.
b)	• Between and within unit inhomogeneity?		C	addresses this clause with the uncertainty budget attached
c)	• Changes in property values during storage?		C	The procedure 7.13 Characterisation and Assignment Of Property Values And Their Uncertainties outlines the contributors listed below and effects of some of the listed contributors are currently studied in the stability monitoring spreadsheet Implementation to be verified during initial assessment
d)	• Changes in property values during transport?			

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CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
e)	NOTE: Where values other that certified values are assigned, a statement of uncertainty could be appropriate and improve the use of the material.			
7.14	**RM documents and labels:**			
7.14.1	• Has the RMP made available a certificate for all CRM's?		C	CRM Name: Certified Reference Material AS007 CRM Code: CRM AS 007 Batch Number: 1 Date of Certification: 21-10-2025 Expiry Date: One year after opening Produced and certified by: Azuresky Pty Ltd, South Africa
	• Has a product information sheet been made available for all other RM?		C	
7.14.2	Do the certificates and product information sheets include at least the following?		C	- Equilibrate to room temperature before use (conditioning), for at least 3 hours. - Determine inherent moisture and adjust dry-basis values to air-dry if required. - Mix the material thoroughly before weighing.
a)	• Title of the document?		C	
b)	• Unique identifier of the RM?		C	
c)	• The name of the RM?		C	CRM Name: Certified Reference Material AS007
d)	• Name and contact details of the RMP?		C	Authorized by: Nonhlanhla Msibi Managing Director, Azuresky Pty Ltd Date: 22-10-2025 Contact: nonhlanhla@azuresky.it.com or info@azuresky.it.com
e)	• intended use		C	This CRM is intended for calibration, validation, control sample and training of staff for the analytical methods listed in section 6 of this document.
f)	• Minimum sample size (where applicable)?		C	Coal sample, ±150g per unit, air-dried, pulverised to <212 µm.

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CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
g)	• Period of validity?		C	Date of Certification: 21-10-2025
h)	• Storage information?		C	Expiry Date: One year after opening Produced and certified by: Azuresky Pty Ltd, South Africa
i)	• Instructions on handling?		C	
j)	• Page number / total number of pages?		C	- Use within 12 months of opening.
k)	• Document version (number)?		C	- Store in a room not exceeding 230C.
l)	• Information on commutability?			Batch Number: 1
	VIM Definition – commutability of a reference material: property of a reference material, demonstrated by the closeness of agreement between the relation among the measured results for a stated quantity in this material, obtained according to two given measurement procedures, and the relation obtained among the measurement results for other specified materials. (Particularly of interest for clinical RM)			
7.13.4	In addition to a) to f) above do the certificates for CRM's also include:			
a)	• A description of the CRM?		C	Indicative Values – Air-Dry Basis (to be used as a guideline)
	• Property of interest, property value and uncertainty?		C	Tolerance values are based on the following: - ISO 1171 reproducibility (ISO Ash): 3% of the mean for ash values > 10%

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CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
	• Measurement procedure for operational defined measurands?		C	- ISO 1928 reproducibility (Calorific value): 0.30 MJ/kg - ISO 562 reproducibility (Volatile matter): 0.5% absolute or 4% of the mean, whichever is greater - ASTM D4239 reproducibility (Total Sulphur): $(0.02+0.09\bar{x})$
	• Metrological traceability of the certified values?		C	
	• Name and function of the person responsible for the approval?		C	Authorized by: Nonhlanhla Msibi Managing Director, Azuresky Pty Ltd Implementation to be verified during initial assessment
7.14.4	• Is the RM label securely attached to individual RM container?		C	The CAB uses durable label stickers from a trusted supplier to securely label its RM containers.
	• Is the design such that it will remain legible for the defined handling and storage conditions?		C	The design is made to remain attached securely to the RM bottle, so it lasts the defined handling and storage conditions
	• Does the label identify the material, the RM, the batch and any other pertinent information, such as a unique sample number?		C	All relevant information pertaining to the CRM produced is addressed visibly on the CRM label
7.14.5	• In cases where size does not permit all required information on the label, is it included elsewhere, and linked by a unique identifier?		C	Any other information relevant to the CRM Is addressed on CRM certificate which is issued on purchase. Implementation to be verified during initial assessment

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Additional / General Comments *This space may also be used to expand on comments in specific sections*

Certified Values – Dry Basis

Measurand	Assigned Value	Uncertainty (±)	Unit	Method
ISO Ash	19.0	±0.46	%	ISO 1171
ISO Volatile Matter	26.8	±0.35	%	ISO 562
Calorific Value	25.72	± 0.16	MJ/kg	ISO 1928
Total Sulphur	0.97	±0.03	%	ASTM D4239
Quick Ash	19.2	±0.47	%	Internal Method

Indicative Values – Air-Dry Basis (to be used as a guideline)
The average moisture obtained from 22 laboratories is 3.70 %

Measurand	Value	Tolerance	Unit	Minimum	Maximum	Method
ISO Ash	18.3	0.55	%	17.8	18.9	ISO 1171
ISO Volatile Matter	25.8	1.03	%	24.8	26.8	ISO 562
Calorific Value	24.77	0.30	MJ/kg	24.47	25.07	ISO 1928
Total Sulphur	0.93	0.10	%	0.83	1.03	ASTM D4239
Quick Ash	18.5	0.56	%	17.9	19.1	Internal Method

Tolerance values are based on the following:
- ISO 1171 reproducibility (ISO Ash): 3% of the mean for ash values > 10%
- ISO 1928 reproducibility (Calorific value): 0.30 MJ/kg
- ISO 562 reproducibility (Volatile matter): 0.5% absolute or 4% of the mean, whichever is greater
- ASTM D4239 reproducibility (Total Sulphur): $(0.02+0.09\overline{x})$

Documents reviewed showed appropriate evidence on the capability on the facility to be considered for an initial assessment to verify the implementation of the methods

Team Leader's Signature:	Neville Tayler	**Date:**	**June 2026**